

THE LLM IN ANGER

WEEK 2, DAY 5

The first call, made robustly. Then streaming, structured output, and knowing what it cost before you spend it.





PART ONE: THE FIRST CALL

Timeouts, retries, and reading what the response actually tells you

YOU ALREADY HAVE RETRIES

YOU JUST DID NOT CHOOSE THEM

THE SDK DEFAULTS, STRAIGHT OUT OF THE BOX

```
max_retries: 2
timeout: Timeout(connect=5.0, read=600, ...)
```

TWO RETRIES YOU NEVER ASKED FOR

Every "one" call is potentially three, with backoff, silently.

A TEN MINUTE READ TIMEOUT

Wire that into an endpoint and one slow response holds a request for 600 seconds.

Neither number is wrong. They are just not yours yet.

NEVER RETURN ONLY THE TEXT

THREE THINGS COME BACK, NOT ONE

On the response	What it tells you
content[0].text	The answer. A list, because tool use and thinking live here too.
usage.input_tokens	What you sent. This is the bill.
usage.output_tokens	What came back. Also the bill, at a higher rate.
stop_reason	Whether you got a whole answer, or it was cut off mid-word.

A truncated answer returns 200 OK. Nothing in the text tells you.

THEIR FAILURE IS NOT YOUR BUG

TRANSLATE THE UPSTREAM INTO YOUR CONTRACT

Upstream raises	You return	Your caller learns
APITimeoutError	504	Something we depend on was too slow
RateLimitError	429	Slow down, try again shortly
APIStatusError	502	Something we depend on is broken
Anything uncaught	500	We have a bug. Usually a lie.

Your caller has never heard of Anthropic. They have heard of you.



PART TWO: IN ANGER

Cost before you spend it, streaming, structured output, and methods that do not exist

COUNT IT BEFORE YOU PAY IT

COUNT_TOKENS COSTS NOTHING

BEFORE THE CALL

```
client.messages.count_tokens(...)
```

A separate, cheap endpoint. Tells you the size of a request without sending it.

AFTER THE CALL

```
response.usage.input_tokens
```

What it actually cost. Compare the two — they should nearly match.

"What does this cost at a thousand firms a day?" is a question you will be asked.

ASK FOR A SHAPE, NOT A PARAGRAPH

YOU ALREADY KNOW THIS TOOL

SAME PYDANTIC MODEL YOU USED ON TUESDAY

```
class FirmAnalysis(BaseModel):  
    tier: str  
    strengths: list[str]  
    risks: list[str]
```

TUESDAY

Validating what arrives from a client you do not control.

TODAY

Validating what arrives from a model you also do not control.

messages.parse, not create. The object lands in parsed_output.

VERIFY THE API EXISTS

FOUR SECONDS, EVERY TIME

BEFORE YOU RUN GENERATED CODE

```
dir(client.messages)
inspect.signature(client.messages.create)
```

Invention	How it fails
A method that does not exist	AttributeError. Loud. You find it in seconds.
A parameter that does not exist	May be quietly ignored. Your setting does nothing.
A response field that does not exist	A plausible number, completely invented. Nobody checks.